Training-Free Keypoint Uncertainty for Lightweight Pose Estimation: Failure Detection and Conformal Error Bounds from One Forward Pass

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Abstract

Pose estimators in rehabilitation, ergonomics or robotics return wrong keypoints without warning, mostly under occlusion. Existing remedies add cost: MC-Dropout repeats inference, and learned uncertainty heads add parameters and training. We propose SpreadCP, a training-free module for coordinate-classification (SimCC) pose estimators. It reads the spread of the coordinate distributions the model already outputs, maps it to a calibrated failure probability, and scales split-conformal error bounds for each keypoint. It adds no parameters and 0.06 ms per person on one CPU thread. With every parameter fitted on a calibration split of COCO val2017 and every value reported on a disjoint test split, SpreadCP detected keypoint failures with AUROC 0.913 and 0.914 on published SimCC models with ViPNAS-MobileNetV3 and ResNet-50, against 0.820 and 0.873 for 20-pass MC-Dropout, which stayed behind at every dropout rate tested. Against a constant radius, its conformal bounds changed the mean radius by -13% on the lightweight model while raising occluded-joint coverage from 0.657 to 0.770. Across three RTMPose models and the heavily occluded OCHuman dataset, it beat MC-Dropout on every model and its bounds kept their coverage while narrowing, but it only matched RTMPose's default confidence at failure detection, and its COCO advantage at flagging occluded joints reversed on OCHuman. A trained uncertainty head did not improve on it: over three seeds, learned σ minus spread was -0.008 in failure AUROC, and the head changed AP by -1.07 points. On a heatmap model the default confidence was as good, so SpreadCP is specific to coordinate classification.

Keywords: human pose estimation, uncertainty estimation, failure detection, conformal prediction, coordinate classification, lightweight models

1. Introduction

A pose estimator that returns a wrong joint with no warning is a problem wherever the output drives a decision, as in rehabilitation, ergonomics or human–robot interaction. Occluded joints are the clearest case. On our test split, the SimCC ViPNAS-MobileNetV3 model places 0.931 of visible joints within the PCK@0.1 tolerance but only 0.663 of occluded ones.

The usual remedies add something to the model. MC-Dropout [1] repeats inference with dropout active, and heteroscedastic heads [2] add parameters and a training run. Recent pose work has moved towards richer probabilistic outputs, including residual likelihoods [3], calibration against object keypoint similarity [4], and probabilistic formulations of the output map [5]. Yet a coordinate-classification head [6] already outputs a full 1D distribution for each axis. On a lightweight model meant for an edge device, every extra pass, parameter or training run counts, so it matters whether that distribution is already enough.

We propose SpreadCP, a training-free uncertainty module for SimCC pose estimators (Section 3). From the two coordinate distributions of each keypoint it computes a spread score, maps the score to a failure probability with two Platt parameters, and scales a split-conformal error bound with it, so every keypoint comes with a radius that contains the true location at a chosen rate. The only fitting is on a small held-out calibration set, and the model itself is not changed. We test it against default confidence, entropy, flip disagreement, 20-pass MC-Dropout and a trained uncertainty head, under an evaluation plan fixed before any result was inspected (Section 4.1).

Our contributions are:

- SpreadCP, a module that turns the output of any SimCC model into calibrated keypoint failure probabilities and conformal error bounds, with no training, no added parameters and a measured overhead of 0.1% of a lightweight model's forward pass on one CPU thread;
- a held-out evaluation on five published SimCC-style models, including three sizes of RTMPose, on COCO and on the heavily occluded OCHuman dataset, in which SpreadCP detects failures better than 20-pass MC-Dropout on every model and dataset (at every dropout rate tested, with and without flip testing, where we varied these); it beats the default confidence of the original SimCC models and matches that of RTMPose, and it flags occluded joints better than the default confidence on COCO but worse on OCHuman;

- a comparison with a trained uncertainty head built for the same purpose (heteroscedastic, calibration and OKS terms, a token mixer and an adaptive target, in a five-arm ablation with three seeds for the main arm), which does not beat SpreadCP and costs AP on the lightweight backbone;
- spread-scaled conformal bounds that keep their coverage on both datasets and are tighter than a constant radius, covering more occluded joints on COCO and the same share at a smaller radius on OCHuman, together with their limits: the marginal guarantee does not reach the occluded subgroup, and the bounds do not help on a heatmap model; and
- a reproducible pipeline in which every number in the paper is generated from result files, and a list of implementation defects we found and fixed (Appendix A).

2. Related work

Pose estimation heads. SimCC [6] treats each coordinate as a classification over sub-pixel bins, and RTMPose [7] builds a real-time pipeline on the same idea. Heatmap models such as SimpleBaseline [8] regress a 2D map. PCT [9] represents a pose as compositional tokens, the inspiration for the token mixer we evaluate. ViPNAS [10] supplies our lightweight backbone, and SOCA-PRNet [11] adds attention to a PoseResNet, drawing on global context [12] and coordinate attention [13].

Uncertainty in pose estimation. RLE [3] learns a residual likelihood for regression. Gu et al. [4] study the mismatch between predicted confidence and OKS, and train a confidence target aligned with OKS, which we reuse as one component of the learned head. ProbPose [5] develops a probabilistic formulation of the keypoint output, and RTMO [14] is a one-stage model with coordinate classification. More recent work covers post-hoc predictive distributions for YOLO-Pose [15], evidential pose estimation [16], and uncertainty-aware models for multi-person [17] and motor-ability-specific [18] settings. These methods add parameters, a training objective or repeated sampling. SpreadCP instead reads the distribution that a SimCC head already produces. Using the dispersion of a predicted distribution as an uncertainty score is a natural idea, and we do not claim it as new in itself. Our contribution is a calibrated, conformal module built on it for SimCC

pose models, and a held-out test of whether costlier machinery improves on it.

Evaluating uncertainty. Sparsification and AUSE come from optical flow [19]. Comparisons of regression uncertainty metrics [20, 21] motivate reporting several metrics, not one. We use Platt scaling [22] and report ECE [23] after calibration. Split conformal prediction [24] has been applied to keypoints for object pose [25]. Benchmark variance [26] motivates our paired intervals and repeated seeds.

3. Proposed method: SpreadCP

SpreadCP sits after a trained SimCC model and before whatever consumes its keypoints. It has three steps, of which only the last two need fitting, and both fit on a held-out calibration set.

Spread score. For keypoint k, a SimCC head outputs logits over W horizontal and H vertical bins. Let p_x and p_y be the distributions the model was trained to output, $\mathrm{softmax}(\beta z)$ of the logits z, where β is the temperature of its training loss ($\beta = 1$ for the original SimCC models, $\beta = 10$ for RTMPose). β is read from the model's definition and is not fitted. Let $b_i = i/r$ be the bin centres in input pixels, with r the split ratio. The score is the joint standard deviation

$$u_k = \sqrt{\mathrm{Var}_{p_x}[b] + \mathrm{Var}_{p_y}[b]}, \qquad \mathrm{Var}_p[b] = \sum_i p(i)\,\big(b_i - \textstyle\sum_j p(j) b_j\big)^2. \quad (1)$$

A sharp, unimodal distribution gives a small u_k. An occluded or ambiguous joint tends to give a wide or multi-modal one, and so a large u_k. The score uses the logits of the pass that produced the keypoint, needs no second pass, and is in pixels, so it can be compared across joints and models.

Calibrated failure probability. A keypoint fails when its error, normalised by the square root of its box area, is at least 0.1 (the complement of PCK@0.1). SpreadCP maps the score to a failure probability with Platt scaling [22] on its logarithm,

$$\hat{P}(\text{fail} \mid u_k) = \mathrm{sigmoid}\big(a \log u_k + c\big), \quad (2)$$

with a and c fitted by logistic regression on the calibration set.

Conformal error bounds. For a target coverage $1 - \alpha$, SpreadCP uses split conformal prediction [24] with the score as a local scale. On n calibration keypoints with normalised errors e_j it takes the ratios e_j/u_j and sets $\hat{q}$ to the $\lceil (n+1)(1-\alpha) \rceil$-th smallest. The region for a new keypoint is a disc of normalised radius $\hat{q}\, u_k$ around the prediction, that is $\hat{q}\, u_k \sqrt{A_{\text{box}}}$ pixels for a box of area A_{box}. If calibration and test keypoints are exchangeable, the region contains the true location with probability at least $1 - \alpha$, marginally over keypoints. Uncertain keypoints get wider regions and confident ones narrower, where a constant radius would treat them alike. We use $1 - \alpha = 0.9$.

Optional instance rescoring. The same score can down-weight uncertain keypoints in the instance score that COCO AP ranks by. We evaluate two such rules (Section 4.7) but do not count them as part of the method, since they did not move AP meaningfully.

Cost. The module adds two softmax operations and two weighted sums per keypoint, plus a sigmoid and a multiplication. It adds no parameters and needs no change to the model or its training. Its measured overhead is in Section 5.7.

4. Experimental setup

4.1. Evaluation plan and data

The evaluation questions, metrics and training arms were written down before the first benchmark run, and later changes were logged with their reasons (`paper/ANALYSIS_PLAN.md`). The plan asked four questions: whether one-pass signals match 20-pass MC-Dropout on pretrained models, whether a trained uncertainty head improves on them, what the added machinery costs, and whether uncertainty-based rescoring raises AP. The paper was framed around SpreadCP after all results were in. That changed the presentation, not any metric, split or comparison. The dropout-rate sweep, the no-flip dumps, the widened rescoring grids and one training arm (Section 4.5) were added after seeing results, and we label them exploratory.

Training uses COCO [27] train2017 only. COCO val2017 is split once at image level, with a fixed seed, into a calibration part of 1,500 images and a test part of 3,500 images. Platt parameters, fusion weights, rescoring parameters and conformal quantiles are fitted on the calibration part. Every reported metric comes from the test part, except the reproduction check in Table 1, which uses all images to match the published protocol. Reported

checkpoints are those of the final epoch, so no checkpoint is chosen using evaluation images.

Keypoint-level metrics use ground-truth boxes, since occlusion flags need a correspondence with annotations. The test part contains 47,149 labelled keypoints from 1,644 images with labelled people: 41,442 visible (v=2) and 5,707 occluded (v=1). Unlabelled keypoints are excluded.

4.2. Models

We use the published MMPose [28] checkpoints of SimCC with ViPNAS-MobileNetV3 (input 256×192), SimCC with ResNet-50, and a SimpleBaseline ResNet-50 heatmap model. Flip testing is on unless stated otherwise. Detector-box AP uses torchvision's Faster R-CNN R50-FPN v2 person boxes. Their person box AP is 59.5 (AP_{50} 86.2), whereas the file behind the published MMPose numbers is named for AP 56.4, so the comparison with published AP in Table 1 is approximate.

4.3. Baselines

SpreadCP's score is compared with every other signal the pipeline can produce. With p_x, p_y and b_i as in Section 3:

- **Peak confidence**: the model's default score, derived from the raw logits. For SimCC it is unbounded above by construction, so it is not a probability.
- **Peak probability**: $\min(\max_i p_x(i), \max_i p_y(i))$.
- **Distribution spread** (SpreadCP's score): u_k from Section 3. On the heatmap model, the analogous spatial standard deviation around a temperature-sharpened soft-argmax, which tests whether the idea carries over to 2D maps.
- **Entropy**: the mean of the two axes' Shannon entropies, each normalised to $[0, 1]$.
- **Flip disagreement**: the distance between the decoded coordinates of the original and flipped views.
- **MC-Dropout**, T=20: the coordinate standard deviation over stochastic passes, with *argmax* and *soft-expectation* decoders, both kept so that the baseline is not handicapped by the decoder choice.
- **Fusion**: logistic regression on peak confidence, spread, entropy and flip disagreement, fitted on the calibration part only.

- **Learned σ and learned OKS**: outputs of the trained head (Section 4.4), reported only for arms that trained them.

MC-Dropout is applied in the *head* only, on the features of one unflipped pass: at the feature layer before the SimCC classifiers (and in the mixer, when present), and as spatial dropout after each deconvolution on the heatmap model. The backbones contain no dropout. The pretrained models were not trained with this dropout, so on them it is a perturbation rather than a trained posterior approximation. The `base` arm in Section 4.5 addresses this by training with the dropout active.

4.4. Trained uncertainty head

The strongest baseline is a head trained for uncertainty on the same backbones. It adds a per-joint log-variance s_k and an OKS logit, with an optional token mixer. With D the per-axis KL divergence summed over bins, the loss adds to SimCC's KL term

$$\mathcal{L}_{\text{het}} = \tfrac{1}{2} e^{-s}\, D + \tfrac{1}{2} s \qquad \text{attenuation [2]}, \tag{3}$$

$$\mathcal{L}_{\text{cal}} = |r|\, e^{-s/2} + \tfrac{1}{2} s \qquad \text{Laplace NLL on the decoded residual } r, \tag{4}$$

$$\mathcal{L}_{\text{oks}} = \text{BCE}\big(z,\ e^{-d^2/(2A(2\kappa)^2)}\big) \qquad \text{OKS target, after [4]}, \tag{5}$$

where r and s are in units of the target width σ_0, d is the decoded error in input pixels, κ the COCO falloff constant of the joint, and A the median train2017 person area in crop pixels. The optional adaptive target is an equal mix of the original label and a Gaussian whose width is scaled by $e^{s/2}$, clamped to $[0.7, 2.5]$, with s detached. Weights are 0.5, 0.2 and 0.2. The token mixer starts as an identity through a zero-initialised residual, so it cannot degrade the pretrained model at the start of training.

4.5. Training arms

Each arm fine-tunes the published checkpoint for 20 epochs on one GPU: AdamW, batch size 64, learning rate 3×10^{-4} for new modules with multipliers 0.1 (backbone) and 0.2 (deconvolution head), cosine decay, mixed precision, and gradient clipping at norm 1. The arms accumulate components: `base` (KL only, the control), `het` ($+\mathcal{L}_{\text{het}}, \mathcal{L}_{\text{cal}}$), `het_oks` ($+\mathcal{L}_{\text{oks}}$), `mix` (+ token mixer) and `full` (+ adaptive target).

One arm is exploratory. At an interim check on all val2017 images, `full` had AP 0.698 at epoch 5 against 0.711 for `base`, with gradient norms of about 100 before clipping. We suspected that e^{-s} in $\mathcal{L}_{\text{het}}$, which grows

as accurate joints reach a low s, multiplies the pose gradient. We kept all planned arms and added `full_sg`, identical to `full` except that D is detached in $\mathcal{L}_{\text{het}}$. Because an evaluation-set number motivated it, its results are reported separately and do not replace `full`.

The SOCA-PRNet comparison trains two backbones from ImageNet weights with the same SimCC head and the KL loss only: a plain ResNet-34 and SOCA-PRNet-34, which is the same ResNet-34 with SOCA attention and, as part of its design, dropout of 0.05 inside the backbone. Both use AdamW with learning rate 5×10^{-4} and weight decay 10^{-4}, batch size 64, a 500-iteration linear warm-up, and 60 epochs with the learning rate divided by 10 after epochs 48 and 57. This is shorter than the 210-epoch schedule of the published models, so neither result can be compared with published numbers. Each backbone was trained once.

4.6. Metrics

A keypoint *fails* when its error, normalised by the square root of its box area, is at least 0.1. Primary metrics are failure AUROC, AUSE on normalised error (retained fractions 0.1 to 1), and occlusion AUROC (v=1 against v=2). AUROC handles ties by average rank, and sparsification breaks ties randomly. Secondary metrics are Spearman ρ with error, ECE after Platt scaling, and the coverage and mean radius of conformal regions at 90% target coverage. Every baseline signal gets the same Platt and conformal treatment as SpreadCP's score, so the comparison is between signals and not between fitting procedures. A constant radius, the conformal region with $u_k = 1$, is the reference for radius.

Intervals are percentile 95% over 1,000 image-level bootstrap resamples, with the same resamples for all signals, so paired differences are valid. We call a difference supported only if its interval excludes zero. AP comes from pycocotools with the MMPose score mode and OKS-NMS. Rescoring differences are the mean paired difference over 10 disjoint folds of the test part.

4.7. Instance rescoring

The default instance score is the box score times the mean confident keypoint score. We compare exponential decay of keypoint scores $c_k e^{-\alpha u_k}$, with u_k normalised by its mean, and an instance score of the box score times the mean closed-form expected OKS under isotropic Gaussian error, $(1 + (\gamma\sigma)^2/(2c))^{-1}$ with $c = A(2\kappa)^2$. Parameters α and γ are selected on the calibration part. Expected OKS is omitted for the heatmap model because its spread is not in pixel units.

Table 1: AP on all val2017 images, with default instance scores. †MMPose model zoo values, not re-measured, obtained with a different detector.

Model	AP (GT boxes)	AP (det. boxes)	AP_{50} (det.)	AR (det.)	Published AP†
SimCC-ViPNAS-MBV3	0.711	0.691	0.882	0.748	0.695
SimCC-ResNet-50	0.739	0.720	0.899	0.774	0.721
Heatmap-ResNet-50	0.737	0.719	0.900	0.768	0.718

4.8. Cost measurement

Latency uses batch size 1, 50 warm-up runs and 300 timed runs, reporting the median, on an RTX 5060 (8 GB) and on one CPU thread of a Core Ultra i7. No training job was running on the GPU, though desktop applications were. FLOPs and parameters come from the unmodified MMPose models where marked "plain". SpreadCP's own overhead is timed the same way on the operations it adds, at the tensor shapes of a 256×192 COCO SimCC model (17 keypoints, 384 and 512 bins), on an otherwise idle GPU. For tensors this small the GPU time is dominated by kernel launches, so it exceeds the CPU time.

5. Results

5.1. Reproduction check

Table 1 shows that our pipeline reproduces published AP to within 0.4 points on detector boxes (0.720 vs 0.721, 0.691 vs 0.695, and 0.719 vs 0.718), despite the different detector.

5.2. Failure detection on published models

Table 2 is the main result, on the published checkpoints with no training. On SimCC, SpreadCP's spread score had the highest failure AUROC and lowest AUSE of all signals on both backbones. Its paired AUROC gain over soft MC-Dropout was +0.093 [0.086, 0.100] on ViPNAS-MobileNetV3 and +0.041 [0.036, 0.047] on ResNet-50. Spread also improved on the default peak confidence of the same model, by +0.019 [0.016, 0.023] and +0.027 [0.022, 0.032], so the default score is not the best one the head already provides. On the heatmap model, the default peak confidence was about level with spread, with both above soft MC-Dropout by +0.079 [0.073, 0.087] and +0.078 [0.071, 0.085]. So the spread's advantage over peak confidence is specific to the SimCC results here, whereas the gap to MC-Dropout holds across all three models. Figure 1 shows the same order across retained fractions. Logistic fusion did not improve on the best single signal in failure AUROC on SimCC, and its intervals overlap those of the best single signal on the heatmap model.

Table 2: Zero-shot uncertainty on the test split, with bootstrap 95% intervals on failure AUROC. Bold marks the best value per column within a model. All values use flip-test predictions, so "1 pass" counts what the signal itself requires: *these need the second, flipped pass.

Signal	Passes	Failure AUROC	AUSE ($\times 10^3$)	Occl. AUROC	Spearman ρ	ECE
SimCC-ViPNAS-MBV3						
Peak confidence	1	0.893 [0.886, 0.900]	8.25	0.780	0.549	0.008
Peak probability	1	0.904 [0.897, 0.912]	6.65	0.797	**0.677**	**0.005**
Distribution spread	1	**0.913** [0.907, 0.919]	**5.90**	0.798	0.660	0.023
Entropy	1	0.909 [0.902, 0.916]	6.33	0.802	0.675	0.013
Flip disagreement	2*	0.817 [0.807, 0.827]	13.10	0.719	0.445	0.016
MC-Dropout, argmax (T=20)	20	0.810 [0.799, 0.820]	13.08	0.744	0.524	0.022
MC-Dropout, soft (T=20)	20	0.820 [0.810, 0.830]	11.22	0.741	0.488	0.020
Fusion (logistic)	2*	0.911 [0.904, 0.918]	6.29	**0.803**	0.664	0.012
SimCC-ResNet-50						
Peak confidence	1	0.887 [0.879, 0.895]	9.16	0.757	0.473	0.011
Peak probability	1	0.899 [0.890, 0.909]	6.44	0.795	**0.683**	**0.005**
Distribution spread	1	**0.914** [0.907, 0.921]	**5.60**	0.801	0.661	0.024
Entropy	1	0.908 [0.899, 0.916]	6.03	0.802	0.680	0.012
Flip disagreement	2*	0.819 [0.808, 0.831]	12.08	0.734	0.460	0.013
MC-Dropout, argmax (T=20)	20	0.828 [0.816, 0.840]	11.53	0.740	0.500	0.036
MC-Dropout, soft (T=20)	20	0.873 [0.863, 0.882]	8.88	0.755	0.521	0.030
Fusion (logistic)	2*	0.912 [0.904, 0.920]	6.03	**0.802**	0.659	0.015
Heatmap-ResNet-50						
Peak confidence	1	0.906 [0.898, 0.914]	6.77	0.807	**0.643**	0.013
Distribution spread	1	0.905 [0.896, 0.913]	7.10	**0.809**	0.628	0.015
Entropy	1	0.900 [0.890, 0.908]	7.46	0.803	0.628	0.023
Flip disagreement	2*	0.762 [0.750, 0.773]	15.83	0.679	0.335	0.050
MC-Dropout, argmax (T=20)	20	0.791 [0.778, 0.802]	14.81	0.705	0.362	0.037
MC-Dropout, soft (T=20)	20	0.827 [0.815, 0.839]	13.76	0.744	0.344	0.029
Fusion (logistic)	2*	**0.907** [0.899, 0.915]	**6.70**	0.807	0.641	**0.004**

Dropout rate. The planned rate of 0.1 could have been a poor choice, so we repeated inference at $p \in \{0.05, 0.2, 0.3, 0.5\}$ (exploratory). The best MC-Dropout result over all rates and both decoders was 0.844 on ViPNAS-MobileNetV3 (argmax, p=0.5), 0.874 on ResNet-50 (soft, p=0.2), and 0.827 on the heatmap model (soft, p=0.1). All are below the corresponding best one-pass signal (Table 3, Figure 2). No rate closes the gap.

Flip testing. Table 2 uses flip-test predictions, so its one-pass signals in fact come from two backbone passes. With flip testing off (exploratory), spread reached 0.912 and 0.912 on SimCC, and peak confidence 0.901 on the heatmap model, against 0.824, 0.877 and 0.825 for soft MC-Dropout, so the one-pass claim holds in the strict sense too.

Occlusion. Occlusion AUROC follows the same pattern, at a lower level. The best signal reached 0.803, 0.802 and 0.809 on the three models, compared with 0.741, 0.755 and 0.744 for soft MC-Dropout. Figure 3 shows instances selected by a fixed rule, not by hand.

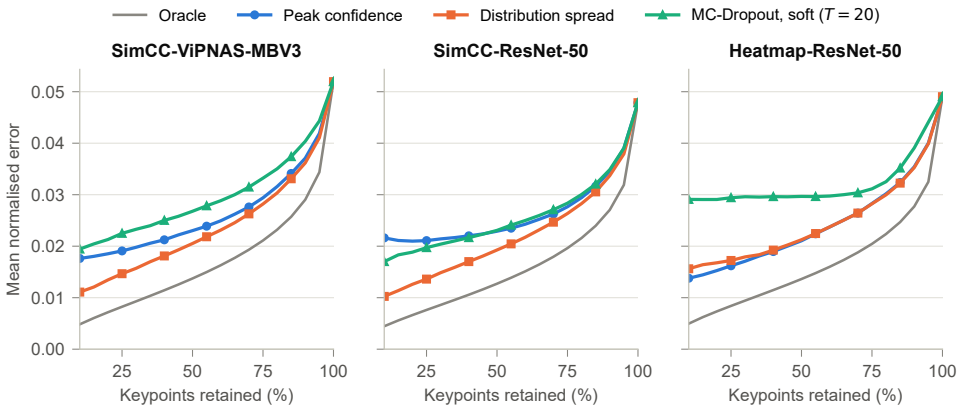


Figure 1: Sparsification on the test split: mean error of the keypoints retained after discarding the most uncertain ones. Lower is better, and the oracle sorts by the true error. Values are in Table 2 (AUSE).

Table 3: Failure AUROC of soft MC-Dropout (T=20) by dropout rate, against one-pass spread. Exploratory. Figure 2 also shows the argmax decoder.

Model	p=0.05	p=0.1	p=0.2	p=0.3	p=0.5	Spread (1 pass)
SimCC-ViPNAS-MBV3	0.817	0.820	0.822	0.823	0.822	0.913
SimCC-ResNet-50	0.873	0.873	0.874	0.873	0.871	0.914
Heatmap-ResNet-50	0.819	0.827	0.812	0.782	0.699	0.905

5.3. Conformal error bounds

Table 4 gives the conformal results, with the same procedure applied to each scale. All variants reach close to the 90% marginal target, as the construction guarantees. A constant radius covered only 0.657 and 0.658 of occluded joints on SimCC. Scaling by spread raised this to 0.770 and 0.762, and changed the mean radius by −13% and −14%. On the heatmap model it raised occluded coverage to 0.875, but at the price of a +47% larger radius, so the benefit there is a trade-off and not a clear gain. Occluded coverage remains below the marginal target in every case: this marginal guarantee does not extend to the occluded subgroup.

5.4. Current lightweight models: RTMPose

RTMPose [7] is the current family of real-time SimCC-style models. We evaluated its published COCO checkpoints at three sizes, from RTMPose-t (3.34 M parameters, 0.36 GFLOPs, 14.4 ms per person on one CPU thread) to RTMPose-m, with the same split, signals and metrics as above and no

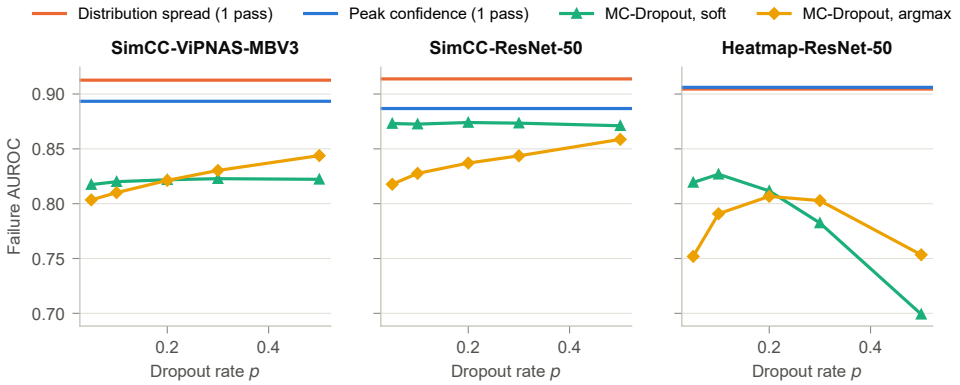


Figure 2: Failure AUROC against dropout rate. One-pass signals are flat lines because they do not depend on the rate.

Table 4: Split-conformal regions at 90% target coverage. Radii are in normalised error units.

Model	Radius scaled by	Coverage	Vis. cov.	Occl. cov.	Mean radius
SimCC-ViPNAS-MBV3	Constant	0.896	0.929	0.657	0.098
	Distribution spread	0.901	0.919	0.770	0.085
	MC-Dropout, soft (T=20)	0.900	0.924	0.719	0.091
SimCC-ResNet-50	Constant	0.898	0.931	0.658	0.091
	Distribution spread	0.899	0.918	0.762	0.078
	MC-Dropout, soft (T=20)	0.897	0.918	0.741	0.085
Heatmap-ResNet-50	Constant	0.896	0.932	0.641	0.090
	Distribution spread	0.902	0.906	0.875	0.132
	MC-Dropout, soft (T=20)	0.900	0.910	0.822	0.135

retraining (Table 5). This evaluation was added after the main study. With detector boxes our pipeline reproduced the published AP to within 0.2 points (0.680, 0.714 and 0.746 against 0.682, 0.716 and 0.746). SpreadCP adds 0.4% to the CPU time of RTMPose-t.

Three findings carry over and one does not. SpreadCP's score again beat 20-pass head MC-Dropout on every model, by +0.017 [0.014, 0.020], +0.013 [0.010, 0.016] and +0.006 [0.003, 0.010] in failure AUROC, though the gap narrows as the model grows. It detected occluded joints better than the default confidence on every model, by +0.036 [0.031, 0.040], +0.038 [0.033, 0.042] and +0.038 [0.033, 0.042] in occlusion AUROC. And its conformal bounds changed the mean radius by −12% to −7% against a constant radius while raising coverage of occluded joints, for RTMPose-t from 0.653 to 0.786. What did not carry over is the gain over the default confidence in failure detection: on RTMPose the two could not be separated (+0.002 [−0.001,

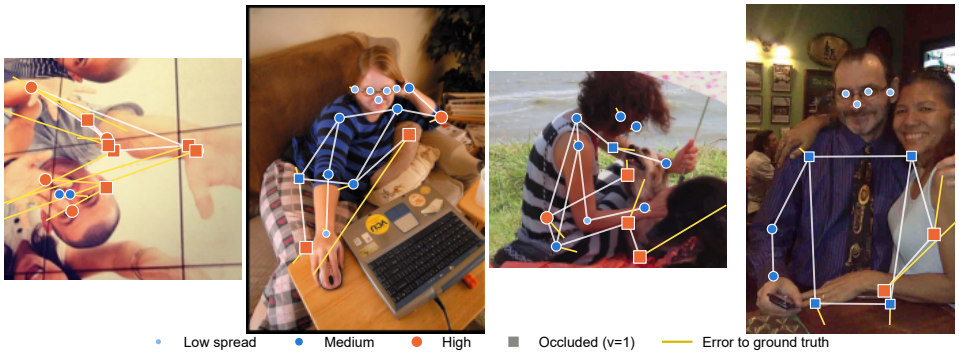


Figure 3: SimCC ViPNAS-MobileNetV3 on test instances chosen by rule: of those with at least 3 occluded and 6 visible labelled joints and a box side of at least 150 pixels, the four with the largest gap between occluded and visible mean error. These are selected failures and do not show typical accuracy.

Table 5: RTMPose with published COCO weights, no retraining. AP is on all val2017 images as a reproduction check; uncertainty metrics are on the test split. "Spread minus" columns are paired failure AUROC differences. Occluded coverage is for 90% conformal regions with a constant radius and with SpreadCP's spread. †MMPose model zoo values with a different detector, not re-measured.

	AP, all val2017			Failure AUROC			Spread minus		Occl. coverage	
Model	GT	det.	Publ.†	Peak conf.	Spread	MC soft	MC soft	Peak conf.	Const.	Spread
RTMPose-t	0.697	0.680	0.682	0.900	0.902	0.886	+0.017	+0.002	0.653	0.786
RTMPose-s	0.733	0.714	0.716	0.906	0.909	0.896	+0.013	+0.003	0.645	0.793
RTMPose-m	0.766	0.746	0.746	0.907	0.910	0.904	+0.006	+0.003	0.656	0.796

0.006], +0.003 [−0.000, 0.006] and +0.003 [−0.001, 0.006]), and the default confidence had the lower AUSE (spread minus confidence +0.93 [0.73, 1.15] on RTMPose-t, $\times 10^{-3}$). RTMPose's loss applies a softmax to its Gaussian targets and a temperature to its logits, which may give its default score a better ranking than the original SimCC models have; we did not test this. The spread was also not the best single signal there: entropy was ahead on RTMPose-t (+0.007 [0.003, 0.011]) and logistic fusion on RTMPose-t and RTMPose-s (+0.009 [0.006, 0.012] and +0.006 [0.002, 0.009]), while the spread led entropy on occlusion on all three. We keep the spread as defined before these models were seen. Rescoring did not help on RTMPose: calibration selected little or no exponential decay (ΔAP −0.01 to +0.00 points with ground-truth boxes), and expected OKS lowered AP by −0.49 to −0.43 and −1.23 to −1.09 points with ground-truth and detector boxes.

5.5. Cross-dataset test under heavy occlusion: OCHuman

OCHuman [29] shows people heavily occluded by other people. We ran the six published COCO models on it without retraining and used its official split: every fitted quantity on OCHuman val (2,500 images) and every reported value on OCHuman test (2,231 images, 54,235 labelled keypoints, of which 9,300 occluded). This evaluation was planned and logged before any OCHuman result was computed. The models are much less accurate here: PCK@0.1 on occluded joints was 0.425 for SimCC-ViPNAS-MobileNetV3 and 0.457 for RTMPose-m, against 0.725 and 0.766 on visible ones. Every signal was weaker as well: the best failure AUROC per model was 0.726 to 0.781, against 0.907 to 0.915 on COCO (Table 6).

Two COCO findings held. SpreadCP's score beat 20-pass MC-Dropout on all 5 SimCC-style models, from +0.019 [0.015, 0.023] on RTMPose-m to +0.059 [0.054, 0.065] on SimCC-ViPNAS-MobileNetV3, and it again beat the default confidence on the original SimCC models (+0.013 [0.010, 0.017] and +0.011 [0.007, 0.014]) while matching it on RTMPose. The conformal bounds kept their coverage under this harder distribution, from 0.896 to 0.902 for every model and scale, and against the constant radius SpreadCP changed the mean radius by −21% to −17% on the SimCC-style models at about the same occluded coverage (for SimCC-ViPNAS-MobileNetV3, 0.848 against 0.850). Here the gain is a tighter bound at equal coverage, where on COCO it was higher occluded coverage.

One finding reversed. On OCHuman the spread detected occluded joints *worse* than the default confidence on every SimCC-style model, by −0.016 [−0.019, −0.012] on SimCC-ViPNAS-MobileNetV3 and −0.058 [−0.061, −0.054] on RTMPose-m, where on COCO it had been better on all five. We did not test why. One possibility is that a joint hidden by another person often produces a sharp distribution on that person's joint, a confident error that a dispersion score cannot see. Occlusion detection should therefore not be read as a general property of SpreadCP. On the heatmap model the conformal radius grew by +81%, as on COCO.

5.6. Comparison with a trained uncertainty head

Could training for uncertainty beat a score that needs no training? This section fine-tunes the baseline head of Section 4.4 on the same backbones and compares its learned outputs with SpreadCP's score computed on the same fine-tuned model, so both see identical weights.

The control, and MC-Dropout after training. The `base` arm fine-tunes the ViPNAS-MobileNetV3 model for 20 epochs with KL only and head dropout

Table 6: OCHuman test, COCO-trained models without retraining, all fitting on OCHuman val. PCK is PCK@0.1 on occluded joints. "Spread minus" columns are paired failure AUROC differences, and the occlusion column the paired occlusion AUROC difference. Occluded coverage is for 90% conformal regions. *Heatmap model, for which the spread is not SpreadCP's SimCC score.

		Failure AUROC			Spread minus		Occl. AUROC	Occl. coverage	
Model	PCK occl.	Peak conf.	Spread	MC soft	MC soft	Peak conf.	spread − conf.	Const.	Spread
SimCC-ViPNAS-MBV3	0.425	0.736	0.750	0.690	+0.059	+0.013	−0.016	0.850	0.848
SimCC-ResNet-50	0.406	0.734	0.744	0.716	+0.029	+0.011	−0.022	0.842	0.837
Heatmap-ResNet-50*	0.422	0.721	0.721	0.658	+0.062	−0.001	−0.005	0.840	0.844
RTMPose-t	0.419	0.770	0.771	0.741	+0.030	+0.001	−0.044	0.840	0.850
RTMPose-s	0.439	0.767	0.765	0.734	+0.031	−0.002	−0.057	0.837	0.843
RTMPose-m	0.457	0.759	0.761	0.743	+0.019	+0.002	−0.058	0.837	0.841

active. Its AP on all val2017 images was 0.714 with ground-truth boxes (pretrained: 0.711) and 0.694 with detector boxes (pretrained: 0.691), so the fine-tuning schedule itself does no harm. Training with dropout did help MC-Dropout: failure AUROC rose from 0.820 to 0.859 for the soft decoder and from 0.810 to 0.835 for argmax, and occlusion AUROC from 0.741 to 0.762. The pretrained comparison in Table 2 is therefore somewhat unfavourable to MC-Dropout. The ranking did not change: spread reached 0.912 on the same model, and soft MC-Dropout minus spread was −0.053 [−0.058, −0.047]. This is one seed on one backbone.

The full learned head. The `full` arm adds the heteroscedastic, calibration and OKS terms, the token mixer and the adaptive target. Table 7 compares it with the control. On ViPNAS-MobileNetV3 with seed 0 (other seeds below), the learned signals did not beat spread computed on the same model's output. Learned σ minus spread was −0.010 [−0.013, −0.007] in failure AUROC, +0.58 [0.41, 0.78] in AUSE ($\times 10^{-3}$, lower is better) and +0.001 [−0.001, 0.003] in occlusion AUROC, so σ was worse on the first two and not distinguishable on the third. The OKS head was further behind, at −0.046 [−0.051, −0.041] in failure AUROC. Adding both to logistic fusion did not separate it from spread alone either (+0.000 [−0.001, 0.002]).

The head also cost accuracy. On val-test, AP with ground-truth boxes was 0.708 against 0.720 for the control, a mean paired fold difference of -1.24 ± 0.56 AP points, positive in 0 of 10 folds. With detector boxes the difference was -1.03 ± 0.40 points. So the deficit seen at the interim check did not close by the final epoch. The median logged gradient norm before clipping, where each logged value is a 50-iteration mean, was 111.3 for `full` and 0.3 for the control, against a clip threshold of 1.0.

Table 7: Fine-tuning arms, final-epoch checkpoints. AP is on val-test images. Δ is the mean paired difference from the control across 10 disjoint val-test folds, in AP points, with the SD across folds, for ground-truth (GT) and detector (det.) boxes. [§]Exploratory arm, added after an interim look at evaluation data (Section 4.5).

		AP, val-test			Failure AUROC			Occlusion AUROC	
Backbone	Arm	GT	Δ GT	Δ det.	Spread	Learned σ	Learned OKS	Spread	Learned σ
ViPNAS-MBV3	Pretrained	0.717	-0.24 ± 0.22	-0.18 ± 0.14	0.913	–	–	0.798	–
	Control (KL only)	0.720	–	–	0.912	–	–	0.799	–
	+ σ head	0.709	-1.09 ± 0.60	-0.84 ± 0.59	0.912	0.907	–	0.799	0.800
	+ OKS head	0.710	-0.94 ± 0.47	-0.86 ± 0.43	0.912	0.907	0.868	0.798	0.799
	+ Token mixer	0.713	-0.75 ± 0.48	-0.54 ± 0.29	0.909	0.903	0.868	0.799	0.800
	+ Adaptive target	0.708	-1.24 ± 0.56	-1.03 ± 0.40	0.913	0.904	0.868	0.796	0.798
	Same, stop-gradient[§]	0.715	-0.38 ± 0.34	-0.40 ± 0.27	0.909	0.905	0.868	0.796	0.799
ResNet-50	Pretrained	0.741	$+0.04 \pm 0.19$	-0.07 ± 0.21	0.914	–	–	0.801	–
	Control (KL only)	0.741	–	–	0.915	–	–	0.801	–
	+ Adaptive target	0.740	-0.26 ± 0.31	-0.15 ± 0.39	0.914	0.898	0.857	0.801	0.796

Stopping the pose gradient (exploratory). The `full_sg` arm detaches D in $\mathcal{L}_{\text{het}}$ and is otherwise identical to `full`. Its logged gradient norm stayed large, with a median of 85.3, so the detach did not remove the large norms that prompted this arm. On val-test its AP was 0.715 with ground-truth boxes, -0.38 ± 0.34 points from the control and $+0.86 \pm 0.48$ from `full` (positive in 10 of 10 folds). With detector boxes these were -0.40 ± 0.27 and $+0.63 \pm 0.42$ (positive in 10 of 10 folds). So on this seed the detach removed most of the AP loss, though not all of it. It did not change the ranking of the signals. Learned σ minus spread was -0.005 [-0.007, -0.002] in failure AUROC and $+0.33$ [0.19, 0.51] in AUSE, and only on occlusion AUROC did σ come out ahead, by $+0.003$ [0.001, 0.004]. That occlusion difference is small. An evaluation-set number prompted this arm and it has one seed, so the AP recovery is a hypothesis for later work and not an established result.

ResNet-50. The larger backbone repeats the signal result but not the size of the AP cost, on one seed. Learned σ minus spread was -0.017 [-0.021, -0.013] in failure AUROC, $+0.71$ [0.55, 0.92] in AUSE and -0.005 [-0.007, -0.002] in occlusion AUROC, so here σ was worse on all three. The OKS head was at -0.057 [-0.064, -0.051] in failure AUROC, and fusion with both learned signals was not separated from spread (-0.001 [-0.003, 0.000]). The AP difference from the control was -0.26 ± 0.31 points with ground-truth boxes and -0.15 ± 0.39 with detector boxes, each smaller than its SD across folds, so we cannot call it a cost on this backbone. Training with dropout again helped MC-Dropout, from 0.873 to 0.882 for the soft decoder, but it stayed behind spread (-0.033 [-0.038, -0.028]).

Which component costs AP. The planned arms on ViPNAS-MobileNetV3 add one component at a time, on one seed. The first step carried most of the

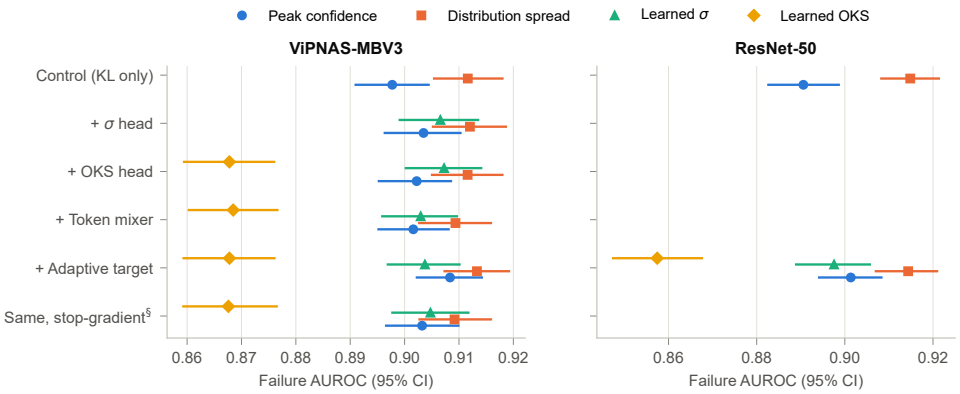


Figure 4: Failure AUROC on val-test for each fine-tuning arm, one seed, with 95% bootstrap intervals. Intervals from different signals on the same images are not independent, so paired differences in the text are the proper comparison. §Exploratory arm.

loss: `het`, which adds only the σ head with $\mathcal{L}_{\text{het}}$ and $\mathcal{L}_{\text{cal}}$, differed from the control by -1.09 ± 0.60 points with ground-truth boxes (positive in 0 of 10 folds) and -0.84 ± 0.59 with detector boxes. Measured directly against the arm before each, with ground-truth boxes, the OKS head changed AP by $+0.14 \pm 0.37$ (positive in 5 of 10 folds), the token mixer by $+0.20 \pm 0.32$ (8 of 10) and the adaptive target by -0.49 ± 0.36 (1 of 10). With detector boxes the adaptive-target step was -0.49 ± 0.46 (1 of 10). So the heteroscedastic and calibration terms are the main source of the cost, the OKS head shows no effect, and the adaptive target may add to it. These later steps are about the size of their fold SD, and folds capture image sampling but not variation between training runs, so they are weak evidence on one seed.

No arm brought learned σ level with spread on failure AUROC. The difference was -0.006 [-0.008, -0.003] for `het`, -0.004 [-0.007, -0.002] for `het_oks` and -0.006 [-0.009, -0.004] for `mix`. On occlusion AUROC these three arms could not be separated from spread ($+0.001$ [-0.001, 0.003], $+0.001$ [-0.001, 0.003] and $+0.000$ [-0.002, 0.002]). The added components did not improve the signal either: σ scored 0.907 in the simplest arm and 0.904 in `full`.

Repeated training runs. We trained `base` and `full` on ViPNAS-MobileNetV3 again with seeds 1 and 2 (Table 8). Each seed's AP difference is the paired fold mean against the control trained with the same seed, so the spread across seeds reflects variation between training runs, which the fold SD does not capture. The AP cost appeared on every seed: -1.07 ± 0.16 points with

Table 8: Seeds 0, 1 and 2 of the control and the full head on ViPNAS-MobileNetV3, mean ± SD across seeds. AP is on val-test with ground-truth boxes. Δ is each seed's mean paired fold difference from the control of the same seed, in AP points, then averaged across seeds. σ and OKS are the learned head's outputs. AUROC values are on val-test.

		AP, val-test, GT		Failure AUROC			Occl. AUROC
Arm	Seeds	AP	Δ	Spread	σ	OKS	Spread
Control (KL only)	3	0.7198 ± 0.0004	–	0.911 ± 0.000	–	–	0.798 ± 0.000
Full head	3	0.7096 ± 0.0017	−1.07 ± 0.16	0.911 ± 0.002	0.903 ± 0.001	0.868 ± 0.001	0.797 ± 0.000

ground-truth boxes (mean and SD over 3 seeds, range −1.24 to −0.91) and −0.86 ± 0.28 with detector boxes (range −1.03 to −0.54). The control itself was stable, at 0.720, 0.720 and 0.719. The signal ranking also held. Learned σ minus spread in failure AUROC was −0.008 ± 0.001 across seeds, with the 95% interval below zero in 3 of 3 seeds. For the OKS head it was −0.043 ± 0.003, below zero in 3 of 3. On occlusion AUROC, σ minus spread averaged +0.001, with the interval above zero in 0 of 3 seeds and below zero in 0 of 3. Three seeds give a rough estimate of run-to-run variation, not a precise one, and the ablation arms, `full_sg` and ResNet-50 remain single runs.

5.7. Cost

SpreadCP adds a median of 0.06 ms per person on one CPU thread and at most 0.21 ms on the GPU, which is 0.1% and 1.9% of a single forward pass of the plain ViPNAS-MobileNetV3 model (49 ms and 10.7 ms). It adds no parameters and no FLOPs that matter at this scale, and no training.

Table 9 gives the cost of the baselines. The trained head adds little in FLOPs: from 0.80 G to 0.82 G on ViPNAS-MobileNetV3 including the mixer. Its parameter overhead matters more on the lightweight model, from 8.65 M to 9.62 M. Head-only MC-Dropout with T=20 costs far less than 20 forward passes because the backbone runs once: 2.7× to 5.2× a single pass on the GPU, and 2.1× to 8.9× on one CPU thread. So the case for SpreadCP over head MC-Dropout rests mainly on its stronger signal and not on latency, and any argument based on an assumed 20× cost overstates the difference. The AP cost of the trained head is reported in Section 5.6.

5.8. Instance rescoring

Can the same score also raise COCO AP? Table 10 shows small AP changes. Exponential decay on spread gained +0.16 and +0.26 AP points with ground-truth boxes, and +0.14 and +0.22 with detector boxes, with fold standard deviations of similar size. Expected OKS changed AP by +0.14 and +0.22 points with ground-truth boxes, and by −0.08 and −0.01 with

Table 9: Model cost at batch size 1, median over 300 runs. "Ratio" is T=20 time over single-pass time on the GPU. ‡Includes the unused uncertainty modules of our head. Sub-10 ms GPU differences should be read with caution, since desktop applications shared the GPU.

Model	Params (M)	FLOPs (G)	GPU (ms) 1 pass	GPU (ms) T=20	Ratio	CPU, 1 thread (ms) 1 pass	CPU, 1 thread (ms) T=20
ViPNAS-MBV3, plain SimCC	8.65	0.80	10.7	–	–	49	–
ViPNAS-MBV3, + σ/OKS head	9.06	0.80	10.2	27.8	2.7×	50	107
ViPNAS-MBV3, + token mixer	9.62	0.82	12.6	41.8	3.3×	50	128
ResNet-50, plain SimCC	36.75	5.50	4.9	–	–	123	–
ResNet-50, + σ/OKS head	37.16	5.50	5.5	21.9	4.0×	122	854
ResNet-50, + token mixer	37.72	5.51	8.1	39.8	4.9×	123	906
ResNet-34, SimCC‡	28.65	4.77	5.0	20.6	4.1×	98	607
SOCA-PRNet-34, SimCC‡	28.95	4.77	6.9	23.9	3.5×	101	606
ResNet-50, heatmap‡	34.30	5.45	5.4	28.0	5.2×	138	1235
RTMPose-t	3.34	0.36	4.6	25.8	5.6×	14	38
RTMPose-s	5.47	0.68	4.9	25.6	5.3×	21	46
RTMPose-m	13.59	1.93	6.7	30.6	4.5×	46	78

detector boxes. On the heatmap model, calibration selected $\alpha = 0$ for both box types, meaning no rescoring. Spread-based rescoring is therefore not a meaningful source of AP on these models. Its value lies in identifying which keypoints to distrust.

The fine-tuned `full` heads add two rules that use the learned outputs, expected OKS from σ and the predicted OKS itself, with parameters again selected on val-cal. On ViPNAS-MobileNetV3 neither helped: expected OKS from σ changed AP by -0.06 and -0.09 points with ground-truth and detector boxes, and the predicted OKS by -0.28 ± 0.16 and -0.30 ± 0.26 (positive in 0 of 10 folds). On ResNet-50, expected OKS from σ gained $+0.27 \pm 0.12$ points with ground-truth boxes and $+0.45 \pm 0.11$ with detector boxes (positive in 10 of 10 folds), the only rescoring gain in this study that clearly exceeds its fold SD. It comes from one training run, among several rules and models, so we treat it as a lead to test with more seeds and not as an established gain. It is also smaller than the AP that the head cost on the lightweight backbone.

5.9. Models trained from ImageNet weights

The published checkpoints were all trained by others. To check that SpreadCP does not depend on them, we trained two SimCC models ourselves from ImageNet weights with the 60-epoch schedule of Section 4.5. The spread score again led: failure AUROC 0.910 on ResNet-34 and 0.913 on SOCA-PRNet-34, against 0.892 and 0.897 for the default peak confidence (Table 11).

Table 10: Instance rescoring on the test split, with parameters selected on the calibration split, for the pretrained SimCC models and the fine-tuned full heads (seed 0). All models use SimCC. ΔAP is in percentage points, paired over 10 folds. Test AP differs from Table 1 because it covers the test images only.

Model	Boxes	Rule	Test AP	ΔAP (mean ± SD, 10 folds)	Folds > 0
ViPNAS-MBV3, pretrained	GT	Default	0.717	–	–
		Exp. decay (spread)	0.719	+0.16 ± 0.17	9/10
		E[OKS] (spread)	0.719	+0.14 ± 0.09	9/10
	Det.	Default	0.698	–	–
		Exp. decay (spread)	0.700	+0.14 ± 0.20	9/10
		E[OKS] (spread)	0.697	−0.08 ± 0.37	5/10
ResNet-50, pretrained	GT	Default	0.741	–	–
		Exp. decay (spread)	0.744	+0.26 ± 0.26	9/10
		E[OKS] (spread)	0.743	+0.22 ± 0.15	10/10
	Det.	Default	0.724	–	–
		Exp. decay (spread)	0.726	+0.22 ± 0.30	8/10
		E[OKS] (spread)	0.724	−0.01 ± 0.30	4/10
ViPNAS-MBV3, full head	GT	Default	0.708	–	–
		Exp. decay (spread)	0.709	+0.08 ± 0.16	9/10
		E[OKS] (spread)	0.707	−0.08 ± 0.09	3/10
		E[OKS] (learned σ)	0.707	−0.06 ± 0.12	4/10
		Learned OKS	0.705	−0.28 ± 0.16	0/10
	Det.	Default	0.690	–	–
		Exp. decay (spread)	0.691	+0.09 ± 0.20	8/10
		E[OKS] (spread)	0.686	−0.37 ± 0.37	1/10
		E[OKS] (learned σ)	0.689	−0.09 ± 0.32	5/10
		Learned OKS	0.687	−0.30 ± 0.26	0/10
ResNet-50, full head	GT	Default	0.740	–	–
		Exp. decay (spread)	0.742	+0.26 ± 0.12	10/10
		E[OKS] (spread)	0.741	+0.12 ± 0.09	10/10
		E[OKS] (learned σ)	0.743	+0.27 ± 0.12	10/10
		Learned OKS	0.741	+0.14 ± 0.10	10/10
	Det.	Default	0.723	–	–
		Exp. decay (spread)	0.725	+0.14 ± 0.13	9/10
		E[OKS] (spread)	0.721	−0.24 ± 0.30	1/10
		E[OKS] (learned σ)	0.728	+0.45 ± 0.11	10/10
		Learned OKS	0.726	+0.23 ± 0.14	9/10

The pair also gives a side result on the backbone itself, since SOCA-PRNet-34 is ResNet-34 with attention added. SOCA-PRNet-34 has 28.95 M parameters against 28.65 M, with FLOPs unchanged at two decimals (4.77 G and 4.77 G). On val-test, AP with ground-truth boxes was 0.715 against 0.712, a mean paired fold difference of +0.44 ± 0.53 points (positive in 9 of 10 folds). With detector boxes it was +0.48 ± 0.56 (positive in 9 of 10 folds). The difference leans towards SOCA-PRNet, but each mean is smaller than its SD across folds, and with one training run per backbone the variation between runs is not measured. Our fine-tuning seeds give only a rough guide to that variation, since training from ImageNet weights for 60 epochs is a different setting. We therefore read this as a small possible gain, not an established one.

Table 11: ResNet-34 and SOCA-PRNet-34 with the same SimCC head, trained from ImageNet weights for 60 epochs, one run each. AP is on val-test images. The shortened schedule means these values cannot be compared with published 210-epoch results.

			AP, val-test		Failure AUROC	
Backbone	Params (M)	FLOPs (G)	GT	det.	Spread	Peak conf.
ResNet-34	28.65	4.77	0.712	0.693	0.910	0.892
SOCA-PRNet-34	28.95	4.77	0.715	0.698	0.913	0.897

6. Discussion

Why the spread score beats head MC-Dropout. We offer explanations, but did not test them causally. The spread summarises ambiguity that the trained classifier has already learned to express: an occluded joint tends to produce a wide or multi-modal distribution. Head-only dropout, by contrast, perturbs a few layers around one fixed backbone feature, and a pretrained model was not fit under this perturbation. Training with the dropout active narrowed the gap but did not close it (Section 5.6), which supports the training part of this account only partly. The remainder may reflect that head dropout around one fixed backbone feature does not reach the sources of error that matter, a possibility we did not test.

Why the learned head did not help. Again we did not test the explanation. A SimCC head is already trained to put probability where the keypoint is, so its spread already carries much of what a separate σ head is asked to learn, and σ is fit to the same training errors. The learned outputs would then mostly restate the spread, which matches how close σ came to it on every arm. The AP cost points to an interaction with the pose objective: the arm that added only $\mathcal{L}_{\text{het}}$ and $\mathcal{L}_{\text{cal}}$ carried most of the cost, and detaching the pose term inside $\mathcal{L}_{\text{het}}$ recovered most of it, though in a single exploratory run. The logged gradient norms stayed large with the detach, so the mechanism is not settled by our data.

Using SpreadCP in practice. A deployed SimCC model already computes the logits SpreadCP needs, so adding it means a few lines of post-processing and one calibration pass over labelled images from the target setting (we used 1,500). The failure probability can flag keypoints for a downstream rule to ignore or for a human to check, and the conformal radius gives a tolerance in pixels that a safety argument can use. Two cautions apply. The coverage guarantee is marginal and holds only when deployment data resemble the calibration data, so the calibration set should come from the setting of use,

and occluded joints were covered below the target in every case. On heatmap models the default score already performs comparably, so SpreadCP offers little there. These findings concern the models and metrics tested, and do not rule out that other methods, such as deep ensembles, would do better at a higher cost.

Limitations. The keypoint-level results use two datasets, COCO and OCHuman, top-down models and ground-truth boxes, so detector errors are not reflected in them. On OCHuman the conformal bounds were calibrated on OCHuman's own val split; we did not test calibrating on one dataset and deploying on another, where the exchangeability behind the guarantee would not hold. MC-Dropout is limited to the head. We did not evaluate full-network dropout or ensembles, and the backbones were not trained with dropout. Only the control and the full head on ViPNAS-MobileNetV3 were trained with three seeds. Every other arm, ResNet-50 and the SOCA comparison are single runs, and all training schedules are short. COCO occlusion flags are annotator judgements. The test split is drawn from val2017, which is a public benchmark and may have influenced the published models. Figures quoted at the third decimal come from one split, though bootstrap intervals capture image sampling within it. Our detector differs from the one behind the published numbers.

7. Conclusion

SpreadCP gives a SimCC pose estimator calibrated keypoint failure probabilities and conformal error bounds without training, without added parameters and at a fraction of a millisecond per person. On five published SimCC-style models, including RTMPose, and on both COCO and the heavily occluded OCHuman dataset, it detected keypoint failures better than 20-pass head MC-Dropout, and its bounds kept their coverage while being tighter than a constant radius. It beat the default confidence at failure detection on the original SimCC models and matched it on RTMPose. It flagged occluded joints better than the default confidence on COCO but worse on OCHuman, where occlusion by other people may cause confident errors that no dispersion score can reveal. It held on models we trained ourselves. A trained uncertainty head, the costlier alternative, did not beat it: the learned σ and OKS outputs ranked failures worse than SpreadCP's score on the same fine-tuned model, on two backbones and, for the lightweight one, on three seeds, and the head cost about one AP point there. Instance rescoring with the score did not move AP meaningfully, and on a heatmap

model SpreadCP offered little over the default score. For SimCC models we would therefore add SpreadCP first, and train an uncertainty head only where a measured gain justifies its parameters and its AP cost.

Appendix A. Defects found and fixed

We report these because each would have changed the results while the code still ran. Regression tests for them are in `tests/test_core.py`.

- **Pretrained heatmap weights failed to load**: inserting dropout as its own layer shifted the deconvolution indices, so those weights were silently left random.
- **Resuming started from random weights**: `-resume auto` cleared the pretrained-checkpoint path even when no checkpoint existed to resume from.
- **The anatomical prior in the mixer was overwritten** by the framework's weight initialisation.
- **ECE was computed on raw logits**, not on probabilities, and on the same data used for fitting.
- **Ties in AUROC and risk-coverage** were ordered arbitrarily, which matters because argmax MC-Dropout often ties at zero.
- **Inconsistent units**: the residual used for calibration and the OKS target were not on the scale their formulas assumed. Unlabelled joints also received an OKS target.

Data and code availability

PENDING, no result yet: Repository URL. COCO is public. Result files regenerate every table, figure and inline number through `scripts/make_paper_assets.py`.

Declaration of competing interest

PENDING, no result yet: To be completed by the authors.

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